Chapter 6: Isometry Group Mn

* Defn 6.1 Let P, Q are points in $\mathbb{R}^n$ and u, v are vector pointing to them respectively.
Distance b/w P and Q $= |\overrightarrow{PQ}| = |v-u|$ (defined in 5.5)

* Defn 6.2 Isometry: $f: \mathbb{R}^n \to \mathbb{R}^n$ (distance preserving maps) s.t if $u, v \in \mathbb{R}^n$, $|f(u) - f(v)| = |u-v|$

* Lemma 6.3 An isometry that fixes the origin $\iff$ Orthogonal linear operator

* Lemma 6.4 All isometries of $\mathbb{R}^n$ together form a group M_n

* Lemma 6.5 If $m \in M_n$, the $\exists$ a unique translation isometry t_a and an orthogonal isometry φ s.t $m = t_a \varphi$

* Lemma 6.6 Let t_a be the translation isometry and φ be the orthogonal linear operator then
$$\varphi t_a = t_{\varphi(a)} \varphi$$

* Lemma 6.7 Group Homomorphism $M_n \xrightarrow{\pi} O_n \xrightarrow{det} \pm 1$

$m \in M_n$: $\det(\pi(m)) = 1$ → Orientation Preserving; $\det(\pi(m)) = -1$ → Orientation Reversing

$\pi(t_a \varphi) \longrightarrow \varphi$

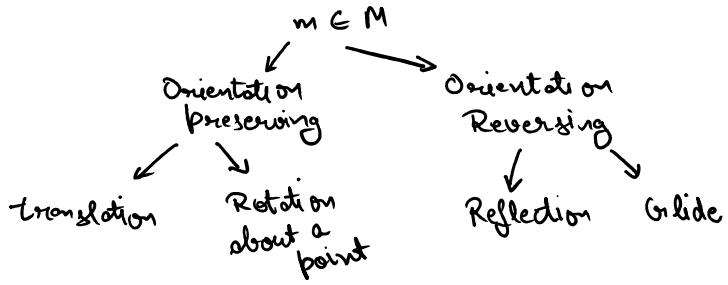
* Lemma 6.8 In $\mathbb{R}^n$, let C and C' be two coordinate systems (Origin + Orthonormal basis) associated with an isometry $n = t_a \varphi$
$T: \mathbb{R}^n \to \mathbb{R}^n$ be an isometry with formula m, m' wrt coordinates C and C' then
$$m' = n^{-1} m n$$

* Defn 6.9 M: Isometry group of $\mathbb{R}^2$
↳ is generated by t_a (translation), ρ_θ (rotation about origin by θ), r (reflection about e_1 axis)

Chapter 6: Finite Subgroups of O2

Saturday, 23 March 2024

* Lemma 6.10



* Defn 6.11 The subgroup Γ of $(\mathbb{R}, +)$ is called discrete if

$$\exists \varepsilon > 0 \text{ s.t } \forall a \in \Gamma \quad abs(a) > \varepsilon$$

* Lemma 6.12 If $\Gamma \subset \mathbb{R}^+$ is discrete then $\Gamma = a\mathbb{Z}$ where $a \in \mathbb{R}$

* Defn 6.13 $G \subset M$ is called discrete if $\exists \varepsilon > 0$ s.t if
  - $t_a \in G$, $a \neq 0$ then $|a| > \varepsilon$
  - $r_\alpha \beta_\theta \in G$, $\theta \neq 0$ then $|\theta| > \varepsilon$

Hint: No need to add a condition specific to $m \in G$ that is a glide or reflection, as m is a translation

* Lemma 6.14 $G \subset O_2$ has finite order $\iff$ $G \subset O_2$ is discrete

- C_n Cyclic group generated by β_θ where $\theta = \frac{2\pi}{n}$, $n \in \mathbb{Z}$
- D_n Dihedral group generated by r and β_θ where $\theta = \frac{2\pi}{n}$, $n \in \mathbb{Z}$

Fixed Point

* Theorem: 6.15 If $G \subset M$ has finite order then G fixes a point and if coordinates are selected appropriately G becomes C_n or D_n

* Defn 6.16 The translation group L of $G \subset M$ is

$$L = \{ v \in \mathbb{R}^2 \mid t_v \in G \}$$

* Lemma 6.17 Let L be a discrete subgroup of the group $(\mathbb{R}^2, +)$ then

a) A bounded region contains only finitely many points

* Lemma 6.17 Let L be a discrete subgroup of the group $(\mathbb{R}^k, +)$ then

a) A bounded region contains only finitely many points

b) If L is non trivial then it contains a non zero vector of minimal length.

Chapter 6: Crystallographic Restriction

Saturday, 23 March 2024

* Lemma 6.18 Let $B=(u,v)$ be the basis of $(\mathbb{R}^2,+)$. Let L be the lattice of integer combinations of B. Then every $v\in(\mathbb{R}^2,+)$ can be uniquely written as $v = x + v_0$ where $x\in L$ and v_0 in $\pi'(B)$.

$\pi(B)$: parallelogram with vertices $(0, u, v, u+v)$ and edges $(u, u+v)$, $(v, u+v)$ missing

* Lemma 6.19 If $L \subset (\mathbb{R}^2,+)$ is discrete

- $L = \{0\}$
- $L = a\mathbb{Z}$ where $a\in\mathbb{R}^2$
- $L = a\mathbb{Z} + b\mathbb{Z}$ where $a, b \in \mathbb{R}^2$

* Defn 6.20 The point group $\bar{G}$ of $G \subset M$ is

$$\bar{G} = \{\varphi\in O_2 \text{ s.t } \exists\, t_a \varphi \in G\}$$

* Theorem 6.21 Let $G\subset M$ is discrete, $a\in L$, $\bar{g}\in\bar{G}$ and $a' = \bar{g}(a) \Rightarrow a'\in L$.

In other words, for a discrete group, its point group maps its translation group to itself

* Lemma 6.22 Crystallographic restrictions :- Let L be a discrete subgroup of $(\mathbb{R}^2,+)$, and let $H\subset O_2$ be a subgroup of the symmetries of L. If L is non trivial, then

* every rotation in H has the order $1,2,3,4$ or 6
* H is one of the group C_n or D_n with $n = 1,2,3,4$ or 6

* Caution : The point group $\bar{G}$ and the translation group L of the discrete subgroup $G\subset M$ doesn't completely identify the group

* Lemma 6.23 The crystallographic restriction limits the types of discrete $G\subset M$ to 17.

* Lemma The concept of crystallographic restriction can be extended to 3D to conclude there

* Lemma 6.24 The concept of crystallographic restriction can be expanded to 3D to conclude there are 230 crystallographic groups

Chapter 6: Group Action

Saturday, 23 March 2024

* **Defn 6.25** Group Operation: Let G be group and S be a set, then the group operation of G on S is map

$$f: G \times S \to S$$

s.t ① $e.s = s \quad \forall s \in S$, e identity elements of G

② $g.(g'.s) = (gg').s \quad \forall g, g' \in G, \forall s \in S$

* **Defn 6.26** Orbit O_s of $s \in S$ is defined as

$$O_s = \{ s' \in S \text{ s.t } g.s = s' \text{ where } g \in G \}$$

* **Defn 6.27** Stabilizer G_s of $s \in S$ is defined as

$$G_s = \{ g \in G \text{ s.t } g.s = s \}$$

* **Defn 6.28** If S consists of only one orbit the group operation is called transitive

* **Lemma 6.29** Let $s \in S$, G operates on S and $H \subset G$ is the stabilizer of s then

① If $a, b \in G$ s.t $a.s = b.s$ then $a^{-1}b \in H$, ie $a = bH$

② If $s' = as$ then the stabilizer of s' is $H' = aHa^{-1}$

* **Lemma 6.30** Let $H \subset G$, then the operation of G on the set of left cosets G/H is transitive $G_{[H]}$ is H. Also if $[x'] \in G/H$ then $G_{[x']} = gHg^{-1}$ where $g \in G$

* **Theorem 6.31** Let S be a set on which G operates, and let $s \in S$ and O_s and H are its orbit and stabilizer. Then there exist a bijective map

$$\varepsilon : G/H \to O_s$$

$$\varepsilon([aH]) = as$$

* **Lemma 6.32** Counting Formula: $|G| = |H|\,|G/H|$

$$= |H|\,|O_s|$$

* **Lemma 6.33** The relation of belonging to an orbit is an equivalence relation $\Rightarrow$ the orbits partition the set S.

* Lemma 6.33 The relation of belonging to an orbit is an equivalence relation $\Rightarrow$ the orbits partition the set S.

$$|S| = |O_1| + |O_2| + \cdots + |O_k|$$

Chapter6: Permutation representation

Saturday, 23 March 2024

* Lemma 6.34 A permutation representation of a group G that operates on a set S is the group homomorphism $\varphi : G \longrightarrow \text{Perm}(S)$

* Defn 6.35 $\text{Perm}(S)$ = Group of permutations of set S

$\text{Perm}(S) \neq$ Permutation representation

* Lemma 6.36 [Operations of G on sets] $\xleftrightarrow[\text{correspondance}]{\text{bijective}}$ [$\varphi : G \longrightarrow \text{Perm}(S)$]

* Defn 6.37 A group operation is called faithful if the only element $g \in G$ st $gs = s \quad \forall s \in S$ is identity, ie the kernel of the permutation representation is $\{e\}$.

* Theorem 6.38 Let G be a finite subgroup of SO_3. Let P represents the set of poles of elements of G. Then G operates on P as a group action

* Lemma 6.39 A finite subgroup of SO_3 is one of the following

① C_k : Cyclic group

② D_k : Dihedral group

③ T : Tetrahedral group of 12 rotational symmetries of tetrahedron

④ O : Octahedral group of 24 rotational symmetries of a cube or a octahedron

⑤ I : Icosahedral group of 60 rotational symmetric of dodecahedron or icosahedron

Hint: Use $G \times P \xrightarrow{\text{group action}} P$ set of poles

and count spins to get

$$\Sigma\left(1 - \frac{1}{r_{ep}}\right) = 2 - \frac{2}{N}$$

* Lemma 6.40 There exists only five regular polyhedra

Chapter7: p Group

Saturday, 23 March 2024

Defn 7.1 Conjugation :- the operation of G on itself is defined by

$$(g, x) \rightsquigarrow gxg^{-1}$$

Defn 7.2 Center of a group G is

$$Z = \{ x \in G \text{ s.t } x.g = g.x \ \forall g \in G \}$$

Defn 7.3 Centralizer of an element x of an group G

$$Z(x) = \{ g \in G \text{ s.t } gx = xg \}$$

Lemma 7.4 $Z(x)$ contains → x, Z, Maybe some other elements

Lemma 7.5

$$|G| = |Z(x)| \ |C(x)|$$

Conjugacy class of x

Lemma 7.6

$$|G| = |C_1| + |C_2| + \dots + |C_k|$$

Lemma 7.7 If $x \in Z$ then it's conjugacy class contains only x itself.

Defn 7.8 p Group :- A group is called p group if its order is positive power of a prime number

Lemma 7.9 The center of a p group is non trivial

Lemma 7.10 Let G, a p group, acts on a finite set S of order n not divisible by p. Then S contains atleast one element which is fixed by all elements of G

***Lemma 7.11** Every group of order p^2 is abelian

*Lemma 7.11 Every group of order p^2 is abelian

* Lemma 7.12 A group of order p^2 is either cyclic or isomorphic to the product of two cyclic groups of order p

Chapter7: Conjugation in Sn

Saturday, 23 March 2024

* Defn 7.13 A group G is called Simple if it has no proper normal groups.

* Lemma 7.14 If a normal group N contains an elements x, then N also contains conjugacy class of x.

* Lemma 7.15 $p, p' \in S_n$ are conjugates if and only if their cycle decompositions have the same order.

* Lemma 7.16 For $n \geq 3$ A_n is generated by 3 cycles

* Lemma 7.17 For $n \geq 5$, the 3 cycles form a single conjugacy class in A_n

* Theorem 7.18 For $n \geq 5$, $A_n \subset S_n$ is a simple group

* Defn 7.19 Let $H \subset G$. Consider conjugation

Orbit of H = conjugate subgroups of form gHg^{-1} where $g \in G$

Stabilizer of H = $N(H) = \{ g \in G \text{ s.t } gHg^{-1} = H \}$

* Lemma 7.20
- $\rightarrow H \triangleleft N \subset G$
- $\rightarrow H \triangleleft G$ if $N(H) = G$

* Lemma 7.21 If $|S| = n = p^e m$, where m is not divisible by p. Then

Let $U = \{$ subset of S of order $p^e \}$

$|U| = N = {}^{n}C_{p^e}$ is not divisible by p

* Theorem 7.22 If G is a group, U is its subset and H is stabilizer of U. Then $|H|$ divides $|G|$ and also $|H|$ divides $|U|$

Hint: $H \times U \longrightarrow U$

Right cosets [$\because$ same size as $|H|$]

$$|U| = |O_{u_1}| + |O_{u_2}| \; __ + |O_{u_k}|$$

Chapter 7: Sylow's p subgroup

Saturday, 23 March 2024

* Defn 7.23 — If $|G| = p^e m$ where $e \geq 1$ and p doesn't divide m. A sylow p subgroup is the subgroup of G of order p^e

* First Theorem 7.24 — If $|G| = p^e m$, where $e \geq 1$ and p doesn't divide m. Then there exists sylow's p subgroups of G.

Hint $G \times S \longrightarrow S$, S: set of subsets of G of order p^e

* Second Theorem 7.25
- If G contains an p subgroup K. Then $K \subset gHg^{-1}$, where H is an sylow's p subgroup
- All the sylow's p subgroups are conjugates of each other

Hint: $G \times G/H \longrightarrow G/H$ (left multiplication, transitive) H: Sylow p group

$c = [H] \in G/H$ with stabilizer H

$K \times G/H \longrightarrow G/H$ K: p group

$|G/H|$ not divisible by $|K|$

$\therefore \exists c' \in G/H$ stabilized by K

Also since $G \times G/H \longrightarrow G/H$ is transitive

$c' = gc$ and $H_{c'} = gHg^{-1}$

$K \subset H_{c'} = gHg^{-1}$

* Third Theorem 7.26 — If s is the number of sylow's p subgroups. Then
1. s divides m
2. $s = kp+1$ where $k \in \mathbb{Z}$

Hint $G \times S \longrightarrow S$ (conjugation, transitive) S: Set of Sylow p subgroup

$[H] \in S$ Sylow p subgroup

$|G| = |N(H)|\,|O_H|$

$= |N(H)|\,|S|$

$\therefore$ s divides m

$H \subset N(H)$

$\therefore p^e \leq |N(H)| \leq p^e m$

$H \times S \longrightarrow S$ (conjugation)

.. s divides m

conjugation
$H \times S \longrightarrow S$

show only $[1] \in S$ has orbit of order 1

Chapter 8: Hermitian Form

Saturday, 23 March 2024

Normal Matrix
$A^*A = AA^*$

* Defn 3.1 Symmetric Matrix
$A^T = A$

Orthogonal Matrix
$A^TA = I$

Hermitian Matrix
$A^* = A$
conjugate transpose

Unitary Matrix
$P^*P = I$

* Defn 3.2 Symmetric Bilinear Form
$\langle , \rangle : V \times V \longrightarrow \mathbb{R}$ s.t
1. Linear in first variable
2. " " second "
3. $\langle v, w \rangle = \langle w, v \rangle \quad \forall v, w \in V$

Hermitian form
$\langle , \rangle : V \times V \longrightarrow \mathbb{C}$
1. Conjugate linear in 1st variable
2. Linear in 2nd variable
3. $\langle v, w \rangle = \overline{\langle w, v \rangle} \quad \forall v, w \in V$

* Defn 3.3
Matrix of form : A where
$a_{ij} = \langle v_i, v_j \rangle$
$[B] = (v_1, v_2, \ldots, v_n)$
basis

* Lemma 3.4 Matrix of the form is Symmetric

Matrix of the form is Hermitian

* Lemma 3.5 $\langle v, w \rangle = X^T A Y$

$\langle v, w \rangle = X^* A Y$

where X and Y are coordinate vectors of v, w w.r.t $[B]$

* Lemma 3.6 If $[B]$ and $[B']$ are two basis of V s.t $[B'] = [B]P$ and A, A' are matrices of $\langle , \rangle$ wrt to $[B]$ and $[B']$ then

$A' = P^TAP$

$A' = P^*AP$

Chapter 8: Orthogonality

Saturday, 23 March 2024

* Lemma 8.1



$\forall v, w \in V$ and $v, w \neq 0_V$

* Defn 8.8 $v \perp w$ if $\langle v, w \rangle = 0$

* Defn 8.9 Orthogonal space $W^\perp$ to subspace $W \subset V$

$$W^\perp = [v \in V \text{ s.t } \langle v, w \rangle = 0_V \ \forall w \in V]$$

* Defn 8.10 $v \in V$ is called a null vector if

$$\langle v, w \rangle = 0 \quad \forall \ w \in V$$

* Defn 8.11 Nullspace N of $\langle\ \rangle$ is the subspace of null vectors.

$$N = [v \mid v \perp V] = V^\perp$$

* Defn 8.12 $\langle , \rangle$ is called non degenerate on V if $N = \{0_V\}$

* Defn 8.13 $\langle , \rangle$ is called non degenerate on $W \subset V$ if $W \cap W^\perp = \{0_V\}$

another way of saying the same thing

if $w \in W$ and $w \neq 0_V$ then $\exists\ w' \in W$ s.t $\langle w, w' \rangle \neq 0$

* Caution A form may be non degenerate on the whole vectorspace and degenerate on subspace and vice versa.

* Lemma 8.14 $\langle , \rangle$ non degenerate $\iff$ matrix of the form A is invertible.

Chapter 8: Orthogonal basis

Saturday, 23 March 2024

* **Theorem 8.15** $\langle , \rangle$ is non degenerate on $W \subset V$ if and only if $V = W \oplus W^{\perp}$

* **Lemma 8.16** If $\langle , \rangle$ not identically zero $\exists\, v \in V$ s.t $\langle v, v \rangle \neq 0$

* **Theorem 8.17** Given $\langle , \rangle$ on V, there exists a orthogonal basis of V wrt $\langle , \rangle$

* **Lemma 8.18** Given $\langle , \rangle$ on V, there exists a basis of V s.t the matrix of the form has this structure

$$A = \begin{bmatrix} I_m & & \\ & -I_n & \\ & & 0 \end{bmatrix}$$

* **Defn 8.19** If $\langle , \rangle$ is non degenerate on $W \subset V$ then $V = W \oplus W^{\perp}$

Orthogonal Projection (linear transformation) $\pi : V \longrightarrow V$ s.t

$\pi(w) = w \quad \forall\, w \in W$

$\pi(w') = 0 \quad \forall\, w' \in W^{\perp}$

* **Lemma 8.20** Projection Formula : If $[v_1, v_2 \cdots v_k]$ is orthogonal basis of W then

$$\pi(v) = c_1 v_1 + c_2 v_2 \cdots + c_k v_k$$

where

$$c_i = \frac{\langle v, w_i \rangle}{\langle w_i, w_i \rangle}$$

* **Defn 8.21** Euclidean Space: Real Vectorspace V + ve definite symmetric bilinear form

* **Defn** Hermitian Space: Complex Vectorspace V

* Def^n 8.22 Hermitian Space: Complex Vector space V + +ve definite Hermitian form

Chapter 8: Normal Operator

Saturday, 23 March 2024

* **Lemma 8.23** If V is a Euclidean or Hermitian space. If $W \subset V$ then the form $\langle , \rangle$ is non degenerate on W and $\therefore$ $V = W \oplus W^{\perp}$

* **Defn 8.24** $T: V \to V$ be a linear operator on a Hermitian space.
  T is called
  normal if its matrix A wrt a orthonormal basis has : $A^* A = A A^*$
  hermitian " " " A " " " " : $A^* = A$
  Unitary " " " A " " " " : $A^* A = I$

* **Lemma 8.25** If A is a normal, hermitian or unitary then $P^* A P$ is also the same, where P is any unitary matrix.

* **Defn 8.26** If $T: V \to V$ is a linear operator on Hermitian space.
  Adjoint operator $T^*: V \to V$ where the matrix of T^* is A^*

* **Lemma 8.27** Let T be a linear operator on Hermitian space and T^* be the adjoint operator.

  $\forall\, v, w \in V$

  (a) $\langle Tv, w \rangle = \langle v, T^* w \rangle$ and $\langle v, Tw \rangle = \langle T^* v, w \rangle$
  (b) if T is normal $\langle Tv, Tw \rangle = \langle T^* v, T^* w \rangle$
  (c) if T is Hermitian $\langle Tv, w \rangle = \langle v, Tw \rangle$
  (d) if T is Unitary $\langle Tv, Tw \rangle = \langle v, w \rangle$

* **Lemma 8.28** Let T be a linear operator on Hermitian space V. Let $W \subset V$
  W is T invariant $\iff$ $W^{\perp}$ is T^* invariant

* Theorem 8.29 If $T: V \to V$ normal operator on a Hermitian Space V.
If (v, λ) is eigen vector of $T \Rightarrow (v, \bar{\lambda})$ is eigenvector of T^*

Chapter 8: Spectral Theorem

Saturday, 23 March 2024

* **Spectral Theorem 8.30**: If $T: V \to V$ is a normal operator on a Hermitian space. There is an orthonormal basis of V with eigen vectors of T.

* **Lemma 8.31**: If A is a normal matrix. There is unitary matrix P s.t P^*AP is diagonal

* **Corollary 8.32**: Every conjugacy class of a normal matrix contains a diagonal matrix

* **Corollary 8.33**: If $T: V \to V$ is an Hermitian operator on a Hermitian space. Then the eigenvalues of T are real.

* **Theorem 8.34**: If $T: V \to V$ is an symmetric operator on a Euclidean space. Then there is a orthonormal basis of V consisting of eigen vectors of T.

* **Lemma 8.35**: If A is a symmetric matrix then $\exists$ a orthonormal matrix P s.t P^TAP is diagonal

* **Defn 8.36**: A bilinear form $\langle , \rangle$ on a vectorspace is called skew-symmetric if

$$\langle v, v \rangle = 0 \quad \forall v \in V \qquad \text{—①}$$

or

$$\langle u, v \rangle = -\langle v, u \rangle \quad \forall u, v \in V \qquad \text{—②}$$

These conditions are same if the characteristic of F is not 2.
If it is 2 condition ① is correct

* **Lemma 8.37**: If $\langle , \rangle$ is a non degenerate skew symmetric form on vector space V the $\exists$ a basis $[B]$ of V s.t the matrix of the form has the following structure.

$$\begin{bmatrix} & \end{bmatrix} \qquad \begin{bmatrix} & \end{bmatrix}$$

2.57 the $\exists$ a basis $[\mathcal{B}]$ of V s.t the matrix of the form has the following structure.

$$S_0 = \begin{bmatrix} \varepsilon & & \\ & \varepsilon & \\ & & \ddots \end{bmatrix} \quad \text{where } \varepsilon = \begin{bmatrix} -1 & \\ & 1 \end{bmatrix}$$